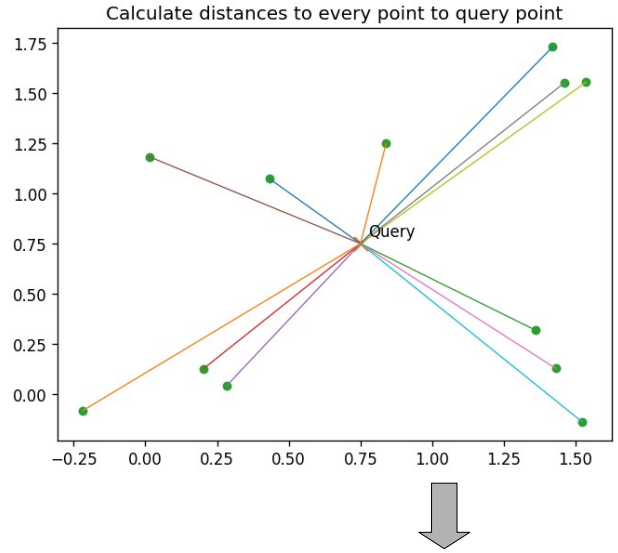
Similarity Search in Vector Databases: kNN, IVF, NSW, HNSW

An overview

K Nearest Neighbors (kNN)

- Finds the k closest points to a query
- By computing distances to all n points
- O(n) per query → slow at scale
- Will find best matches

kNN- algorithm



Distance
1.275413
0.847395
0.829449
0.746751
0.921099
1.174519
1.072903
1.187165
1.125681
0.454620
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Distances

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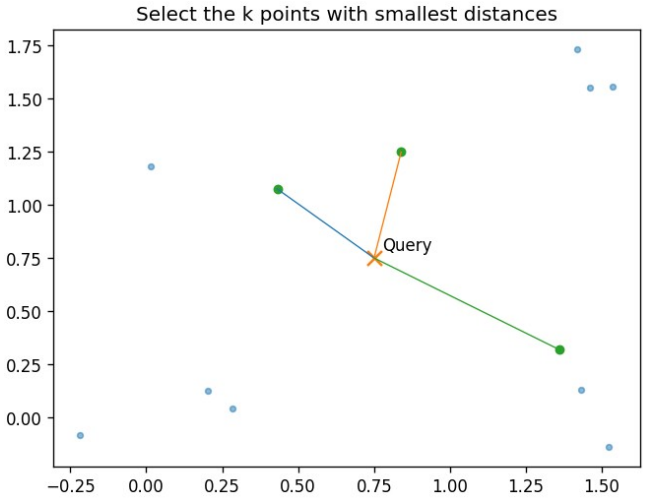
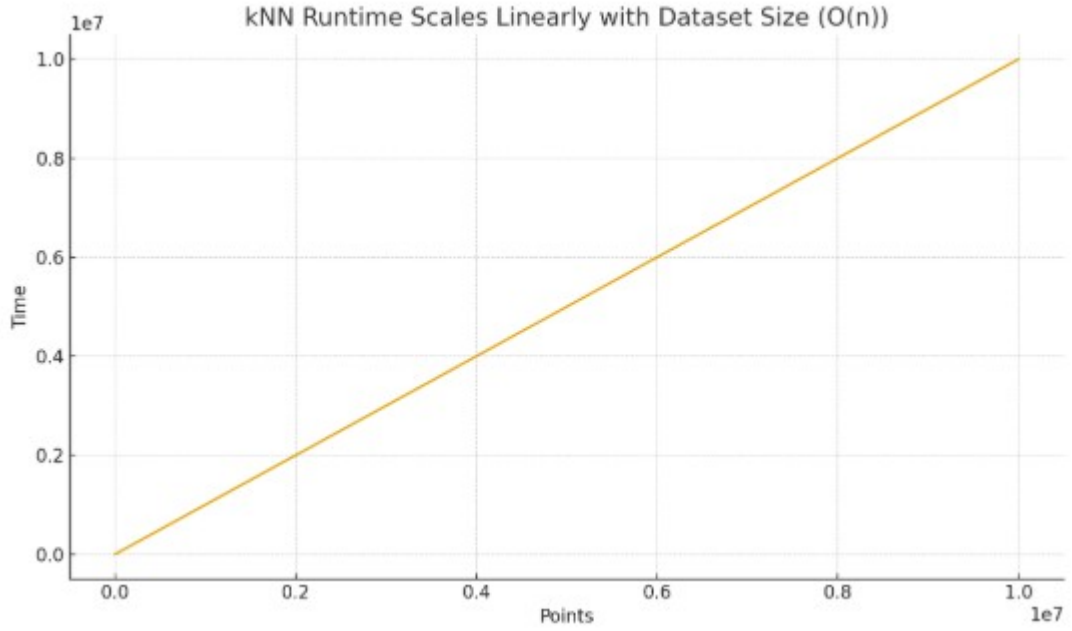
Sorted Distances

Select k closest



Why kNN Doesn’t Scale

- Per-query time grows linearly with data size: kNN must compare a query to every point → O(n) work per query
- latency explodes as n grows.

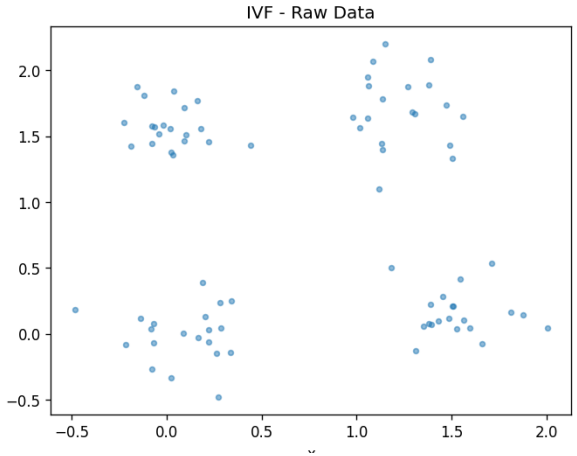


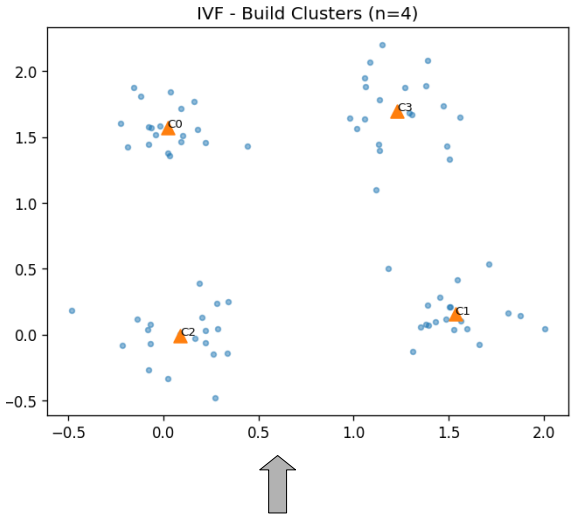
IVF: Inverted File Index — Idea

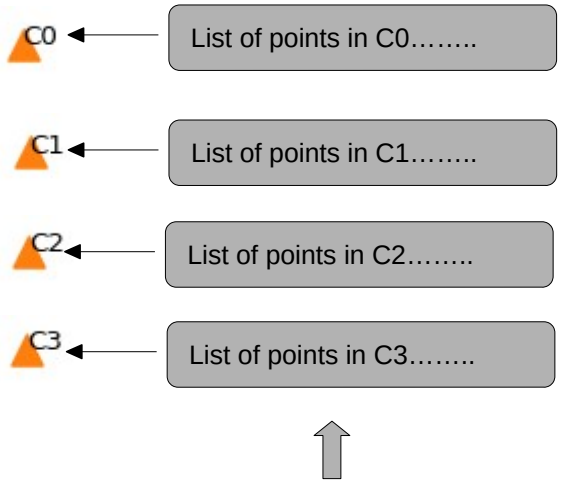
- Partition space into n clusters (n_probe)
  - Rule of thumb: 1000 clusters for 10000000 points -
- At query time probe only k closest clusters. Pick closest point(s) in each.
- So if k=2 and have 1000 clusters, then just search 2 out of 1000 for relevant points

IVF: Build Clusters

Partition space into n_probe=4 clusters, build list of points in each cluster (n_clusters is a hyperparameter)







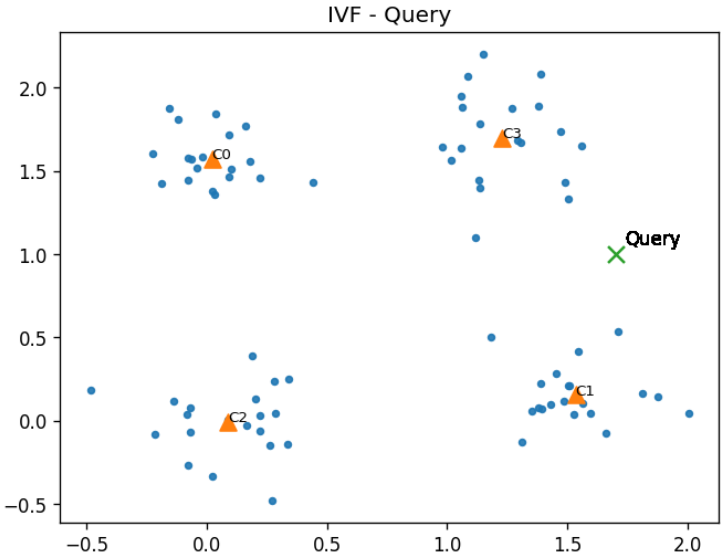
This is an iterative process (KMeans)

1) Randomly place 4 centroids
2) Assign each point to its closest centroid
3) Move centroid to center of its points
4) Go to 2 until no points change cluster

Have list of all points in each cluster

IVF: Query time

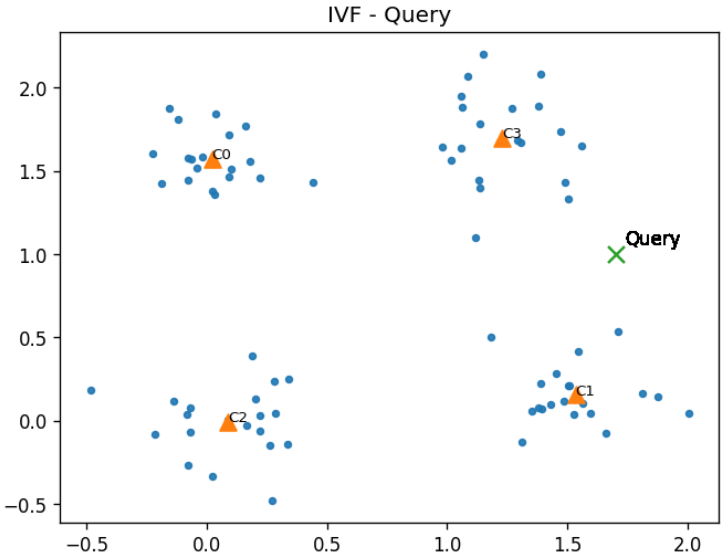
Ex. Find closest 3 points to Query



IVF: Query time

Ex. Find closest 3 points to Query

Pick the nearest cluster to query and search there?



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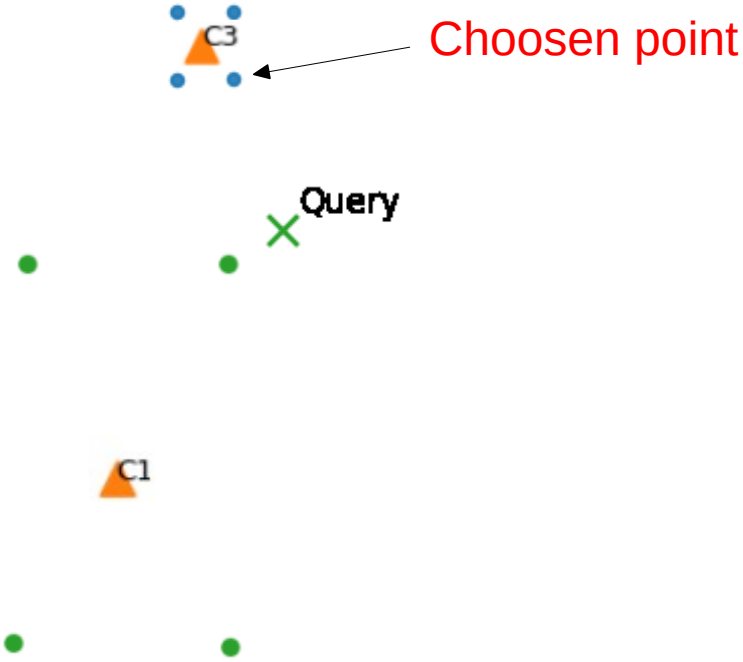
Query is closest to C3, so lower right blue point is marked as closest



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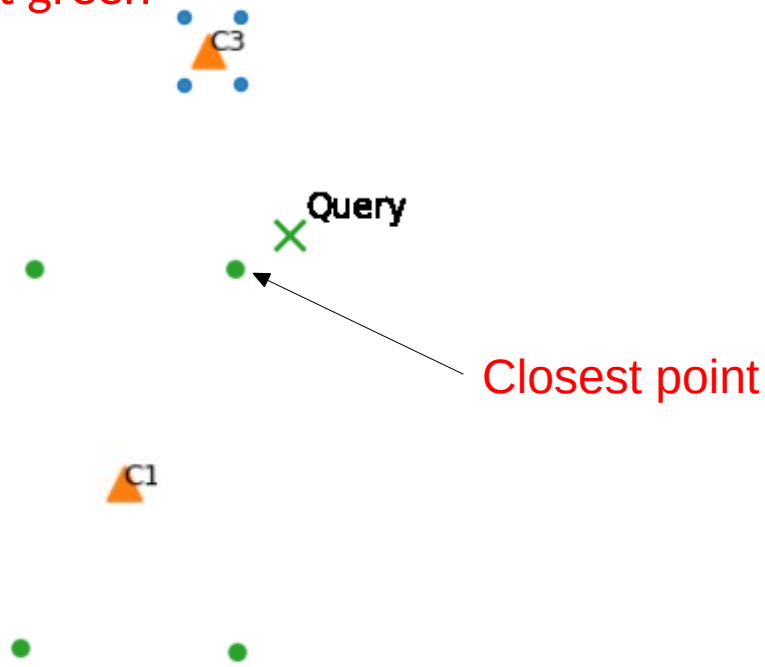

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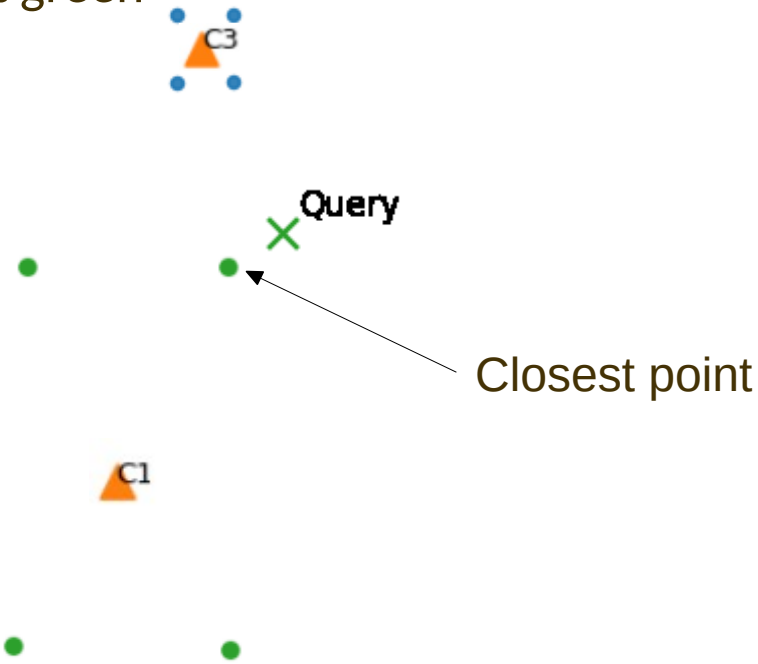
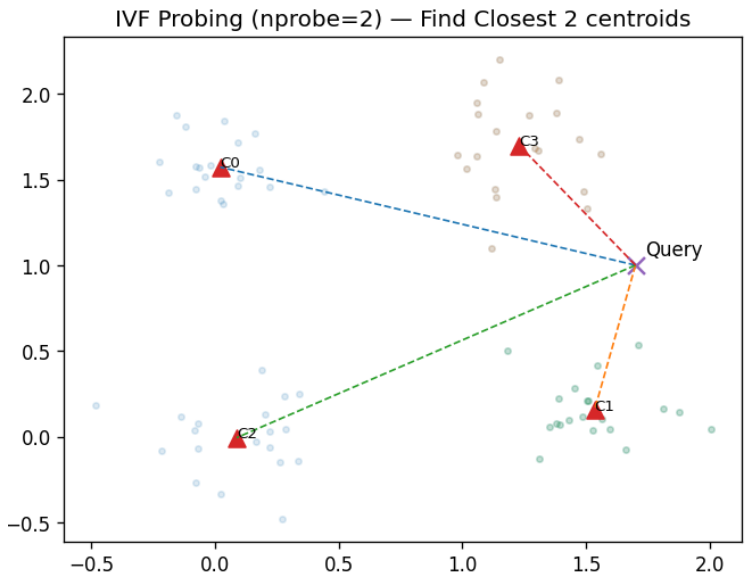


Partial solution is to allow searching more than 1 cluster

IVF: Query time

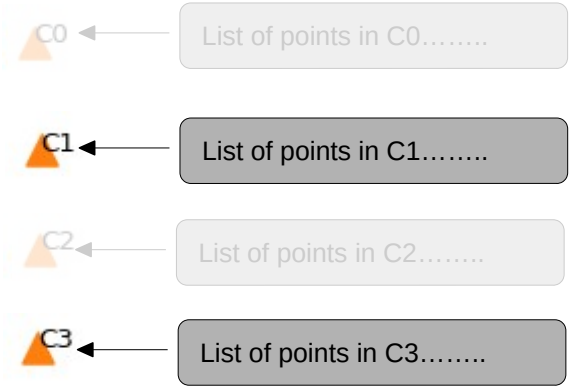
Ex. Find closest 3 points to Query

Find Closest **2** clusters



C1 and C3 are closest centroids

Use the Lists for C1 and C3 to get all the points to compare to

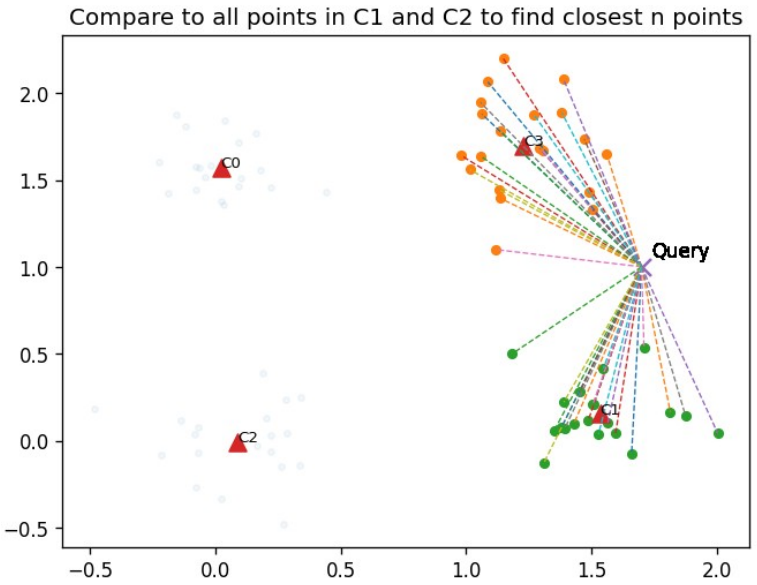


IVF: Query time

Ex. Find closest 3 points to Query

Find Closest 2 clusters

Calculate distances to all points in each cluster



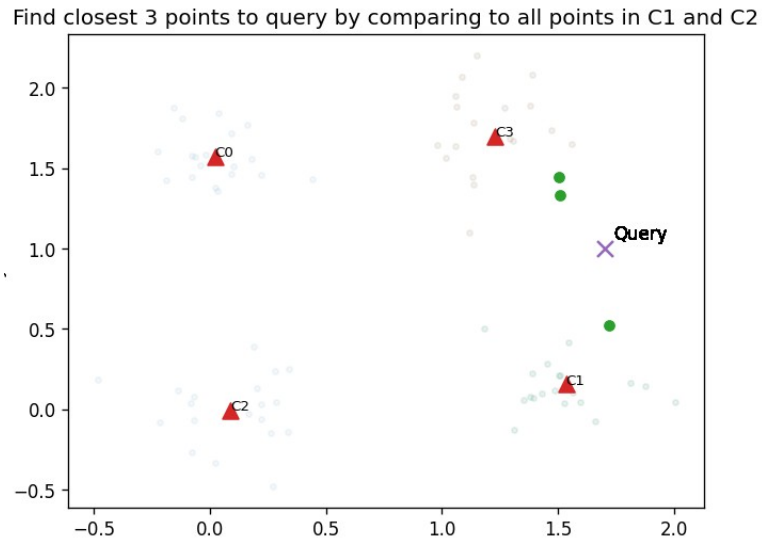
IVF: Query time

Ex. Find closest 3 points to Query

Find Closest 2 clusters

Calculate distances to all points in each cluster

Sort distances,Select closest 3 points



NSW: Navigable Small Worlds — Idea

- Create a connected graph of all points
- Connections are made from each point to other points via ‘nearest neighbor’ algorithm. Connected points are called ‘friends’
- Limit the number of friends a node can have
- Each point keeps a list of its friends

NSW: Navigable Small Worlds — Search

1. Start at pre-defined entry point
2. Greedy Search: choose path to friend that is closest to query
3. If no friends closer than current node we are done (a local minima)
4. Go to 2

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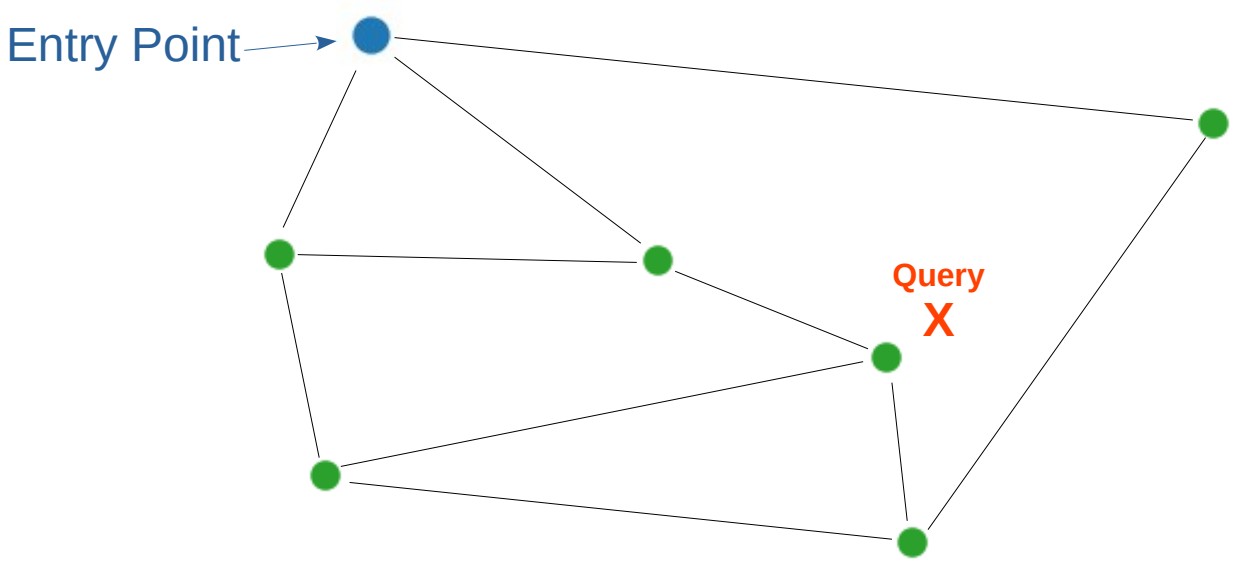

See https://www.pinecone.io/learn/series/faiss/hnsw/ for original image

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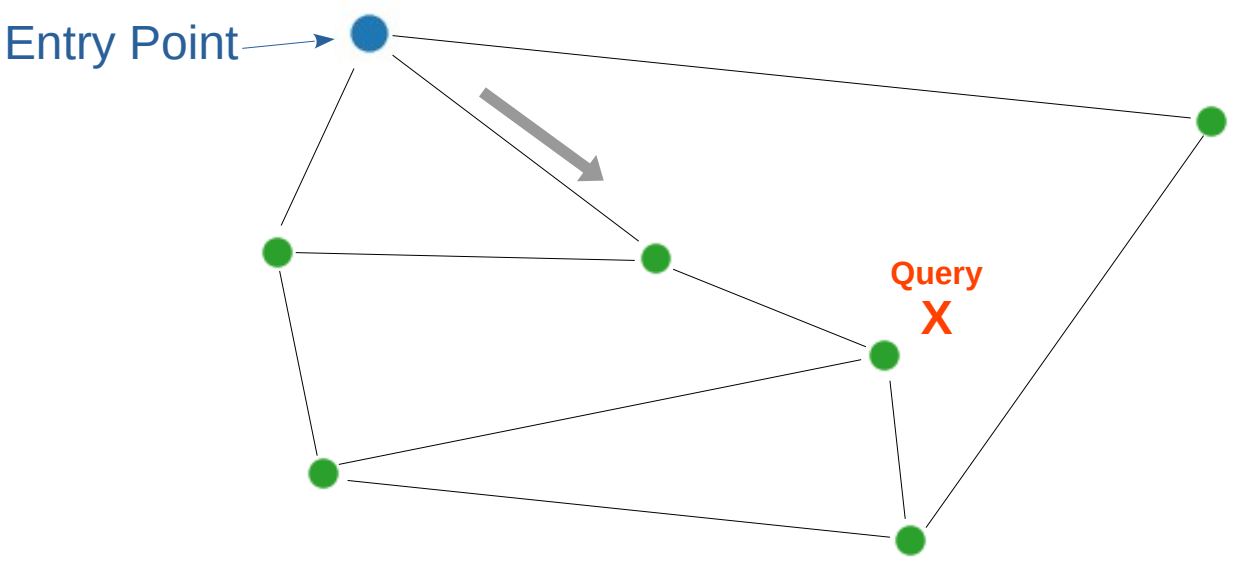

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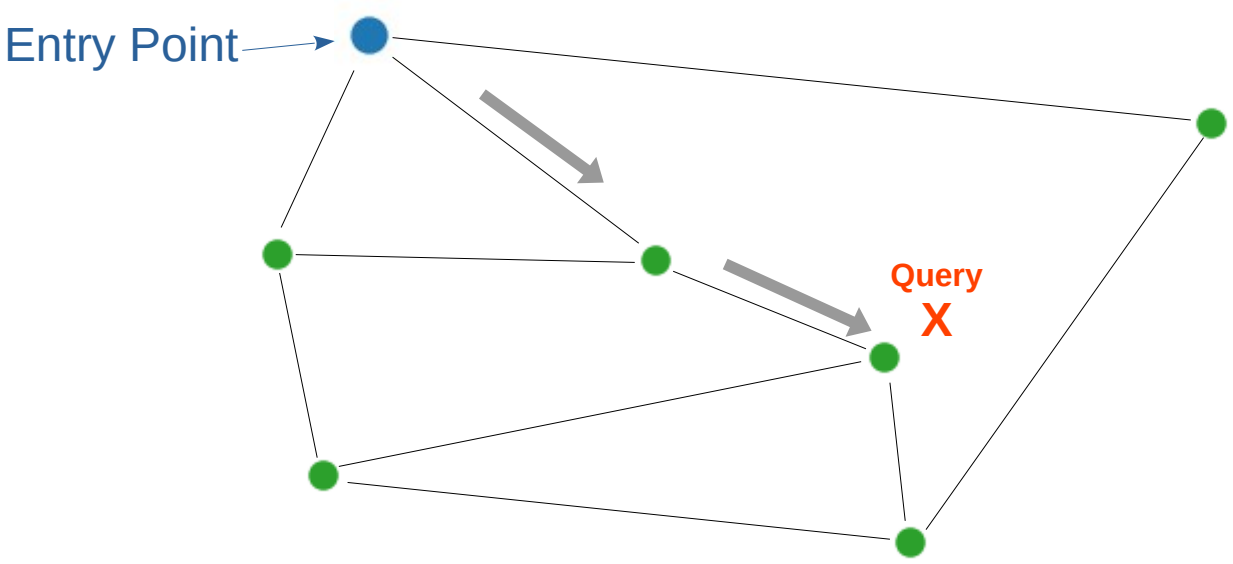

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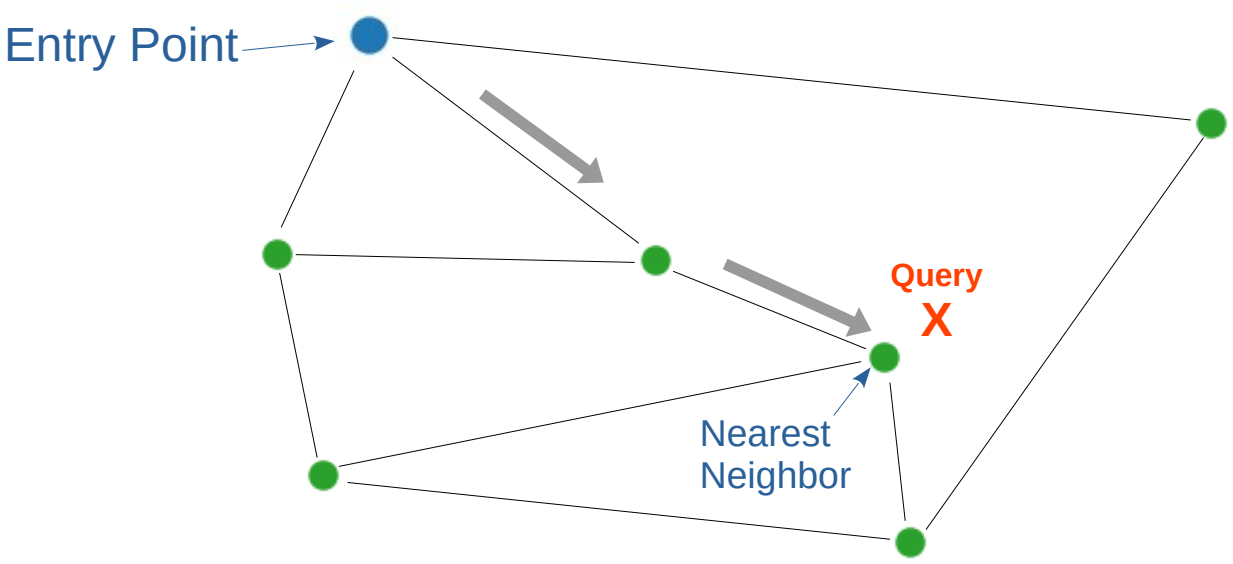

NSW: Navigable Small Worlds — Problem

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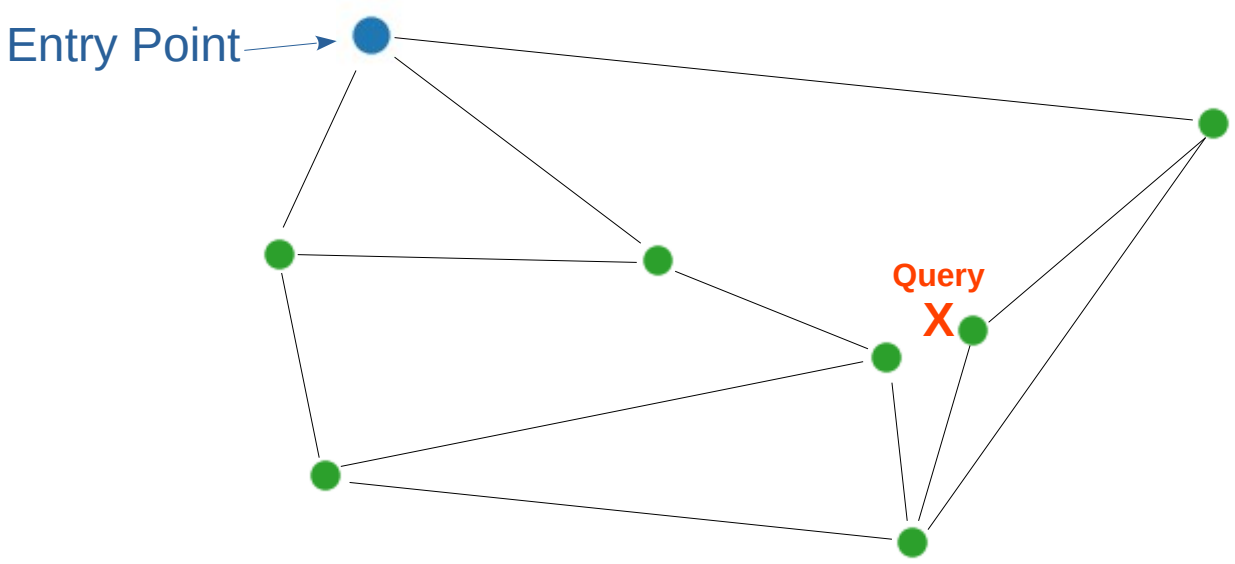

But NSW does not guarantee correct nearest neighbor, add 1 more node

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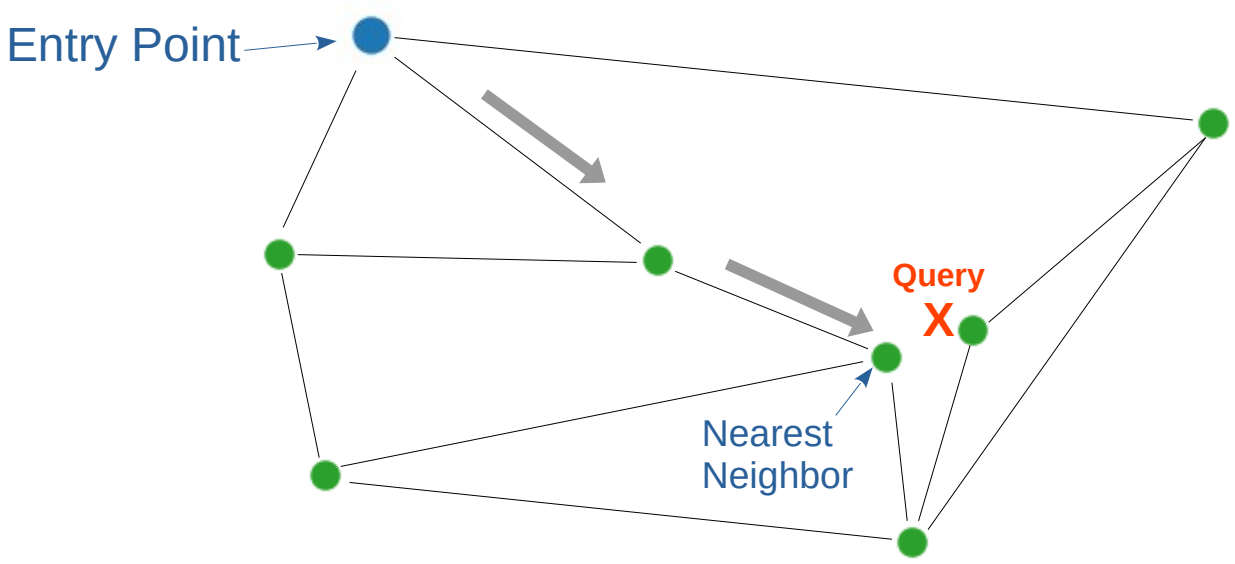
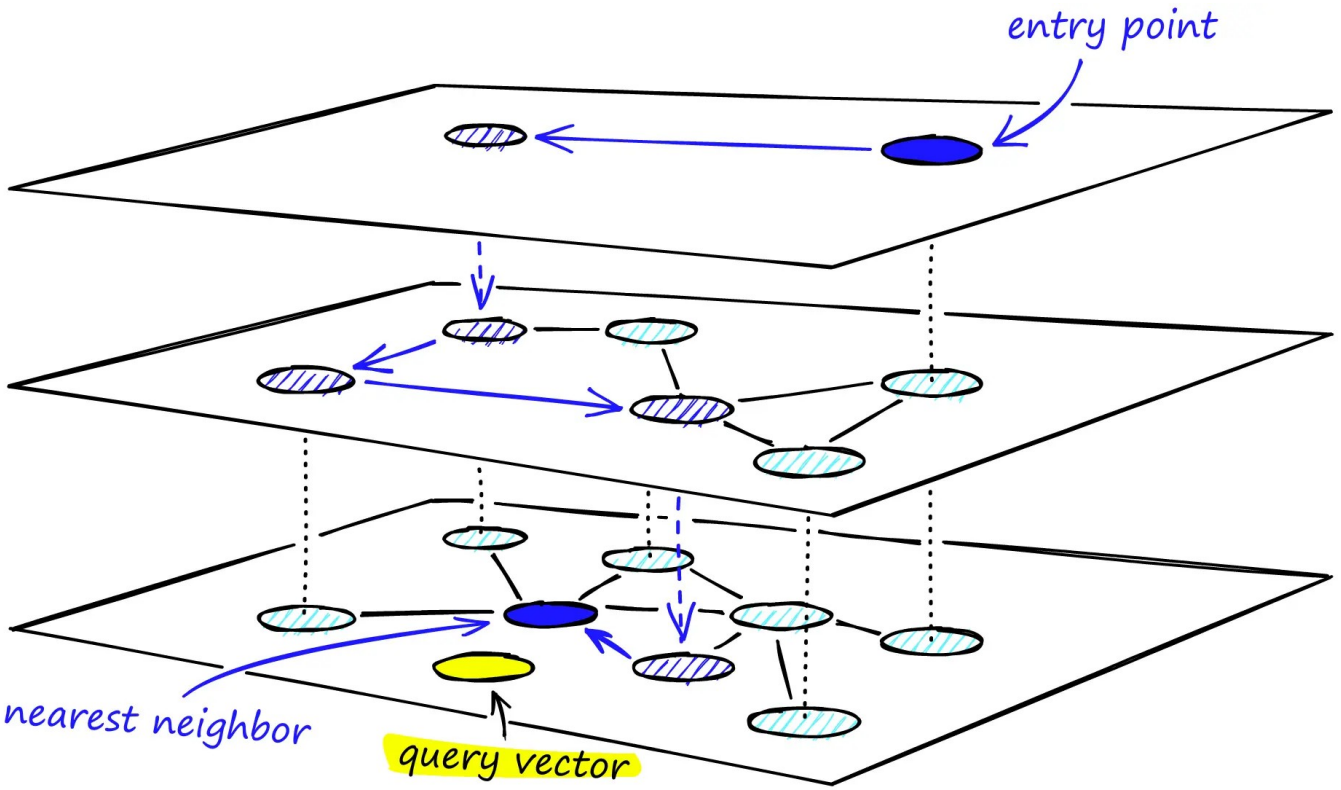


But NSW does not guarantee correct nearest neighbor, add 1 more node

Algorithm never selects best node, NSW is an approximation

HNSW: Heiarchial Navigable Small Worlds



- Multi-layer graph; greedy search from top layer, descending through denser layers.
- Evolution of NSW

HNSW: Heiarchial Navigable Small Worlds-Build

SIMPLIFIED OVERVIEW

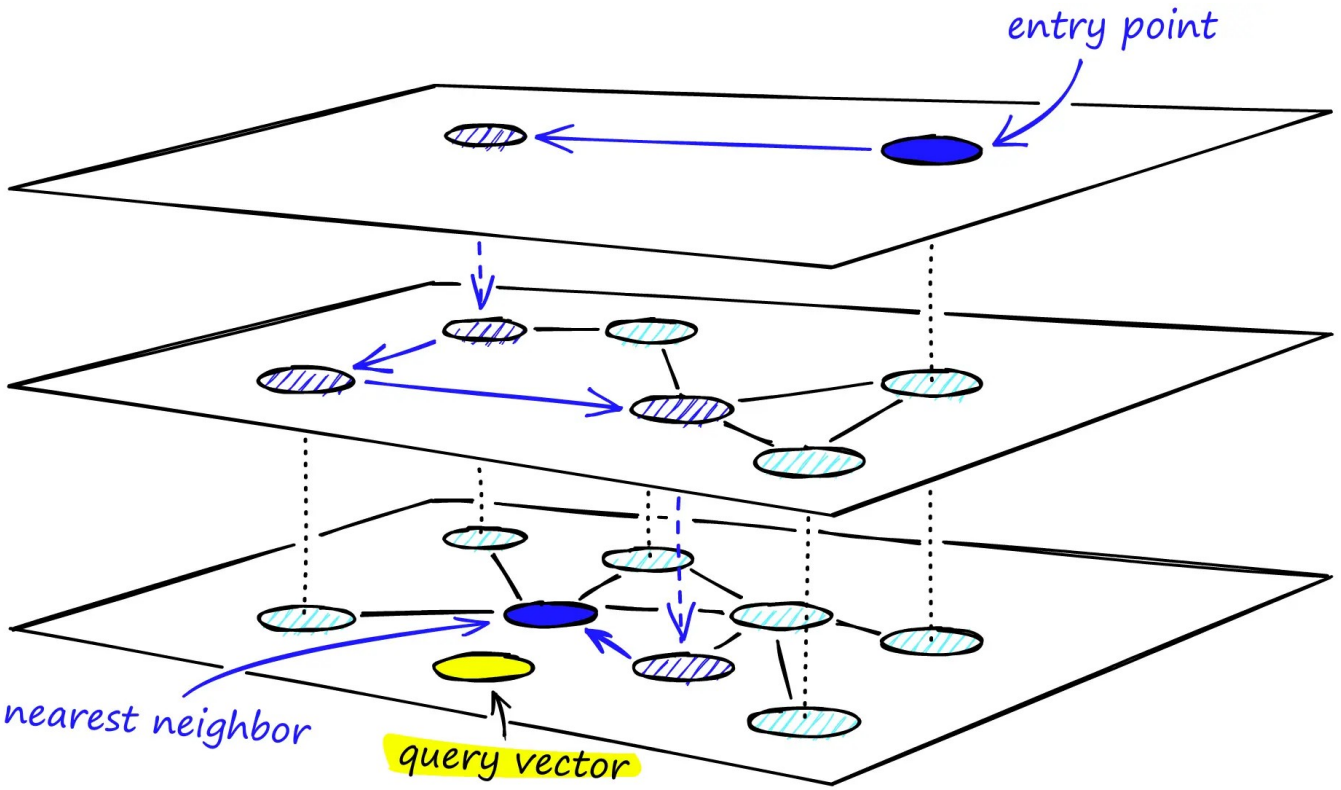
1. Start with single layer, a NSW graph with all points.
2. Generate a decreasing list of probabilities prob[numb_layers] (1 per layer)

   ex. prob=[.97, .025, .005]
3. Cur_layer =0
4. Nex_layer=Cur_layer+1
5. Use probabilistic insertion for all points in layer

   ex. pt1= np.random.rand() #for layer 1 if pt1<prob[1] pt1 inserted in layer 1
6. Get a smaller, distributed representation of lower layer in new layer
7. Build friend list for this layer
8. Stop when have enough layers, otherwise go to 4

Define entry point at top layer

HNSW: Heiarchial Navigable Small Worlds-Use



1. Start at entry
2. Traverse edges in each layer, greedily moving to the nearest vertex to Query (local minima)
3. Drop down a layer.
4. Go to step 2 unless you are on bottom layer (layer 0)

See https://www.pinecone.io/learn/series/faiss/hnsw/ for original image

Quick Comparison

Aspect	kNN (Exact)	IVF	HNSW
Query time	Highest (O(n))	Low, tunable (nprobe - #clusters)	Very low, tunable (efSearch*)
Build time	None	Moderate (k-means)	High (graph building)
Memory	Data only	Data + centroids	Data + graph links (higher)

- kNN will find exact match
- IVF and HNSW may not

*Efsearch controls greedyness of search, higher number means search more candidates per node

Summary

- kNN is exact but slow for large n
- IVF: partition + probe
- NSW: connected graph with greedy search
- HNSW: layered graph, fast greedy search